

Solutions: Linear Algebra

ECON 441: Introduction to Mathematical Economics

Instructor: Div Bhagia

Exercise 4.2

$$1. \ A = \begin{bmatrix} 7 & -1 \\ 6 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 4 \\ 3 & -2 \end{bmatrix} \quad C = \begin{bmatrix} 8 & 3 \\ 6 & 1 \end{bmatrix}$$

$$(a) \ A + B = \begin{bmatrix} 7 & 3 \\ 9 & 7 \end{bmatrix}$$

$$(b) \ C - A = \begin{bmatrix} 1 & 4 \\ 0 & -8 \end{bmatrix}$$

$$(c) \ 3A = \begin{bmatrix} 21 & -3 \\ 18 & 27 \end{bmatrix}$$

$$(d) \ 4B + 2C = \begin{bmatrix} 0 & 16 \\ 12 & -8 \end{bmatrix} + \begin{bmatrix} 16 & 6 \\ 12 & 2 \end{bmatrix} = \begin{bmatrix} 16 & 22 \\ 24 & -6 \end{bmatrix}$$

$$2. \ A = \begin{bmatrix} 2 & 8 \\ 3 & 0 \\ 5 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix} \quad C = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$$

- (a) AB is defined as number of columns in A is two which is equal to the number of rows in B .

$$AB = \begin{bmatrix} 2 \times 2 + 8 \times 3 & 2 \times 0 + 8 \times 8 \\ 3 \times 2 + 0 \times 3 & 3 \times 0 + 0 \times 8 \\ 5 \times 2 + 1 \times 3 & 5 \times 0 + 1 \times 8 \end{bmatrix} = \begin{bmatrix} 28 & 64 \\ 6 & 0 \\ 13 & 8 \end{bmatrix}$$

Not possible to calculate BA as B has two columns, but A has three rows.

- (b) BC and CB are both defined as both have two rows and two columns.

$$BC = \begin{bmatrix} 14 & 4 \\ 69 & 30 \end{bmatrix} \neq CB = \begin{bmatrix} 20 & 16 \\ 21 & 24 \end{bmatrix}$$

4.

$$(a) \begin{bmatrix} 0 & 2 \\ 36 & 20 \\ 16 & 3 \end{bmatrix}_{3 \times 2}$$

$$(b) \begin{bmatrix} 49 & 3 \\ 4 & 3 \end{bmatrix}_{2 \times 2}$$

$$(c) \begin{bmatrix} 3x + 5y \\ 4x + 2y - 7z \end{bmatrix}_{2 \times 1}$$

$$(d) [7a + c \quad 2b + 4c]$$